

CO2-AT1 – Scenario-Based Requirement Analysis

Problem Statement: Smart Grid Cybersecurity Management Platform

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1. Scenario Study and Analysis

1.1 Business Domain

The modern electrical power ecosystem is shifting from a passive, one-directional distribution network into an intelligent, bidirectional Smart Grid, integrating sensing, communication, and automation for real-time monitoring, dynamic load balancing, and self-healing. Substations and feeder lines now carry intelligent electronic devices (IEDs) linked to SCADA systems that give operators real-time visibility into voltage, load, and equipment health while issuing commands to RTUs and PLCs. A defining feature is the proliferation of IoT devices: smart meters report consumption for dynamic pricing and outage detection, while sensors on transformers and renewable assets feed predictive-maintenance analytics — expanding the attack surface well beyond legacy, air-gapped systems and making cybersecurity in Smart Grids a critical discipline.

The convergence of Operational Technology and IT means vulnerabilities can propagate into physical control systems, risking cascading blackouts, and utilities face mounting pressure to adopt layered defenses. Growing renewable integration adds inverter control systems and third-party monitoring endpoints as further entry points. Artificial Intelligence has emerged as a transformative response, analyzing network traffic and device telemetry at grid scale to detect subtle anomalies and enable earlier detection of reconnaissance or insider misuse. It is within this domain — power engineering, industrial control systems,.

1.2 Problem Description

Utility companies currently rely on a fragmented set of monitoring tools — network monitoring, SCADA diagnostics, meter data management, and security alerting handled by loosely integrated systems — making it hard to correlate events and respond before incidents escalate. The proposed platform consolidates these into a single AI-augmented system that continuously monitors smart meters, substation controllers, IEDs, and DER inverters, and must detect cyber attacks such as denial-of-service, command injection, and unauthorized configuration changes, alongside malware and intrusion indicators across both IT and OT segments.

Substations require dedicated physical-and-cyber monitoring, while the far larger population of smart meters needs lightweight mechanisms to detect tampering, cloning, and firmware compromise. AI-based threat prediction and risk scoring must anticipate and

prioritize attacks, with real-time alerts reaching stakeholders through a unified dashboard and automated response for authorized containment. The platform must additionally support forensic log analysis, vulnerability assessment, and continuous compliance monitoring — moving smart grid security from reactive and siloed to proactive and intelligence-driven.

1.3 Existing Challenges

A review of current smart grid security operations reveals a number of significant, persistent challenges that the proposed platform must directly address:

- **Manual Monitoring Dependency:** Many utilities still rely heavily on manual review of logs and dashboards, which is labour-intensive, error-prone, and unable to scale with the volume of telemetry generated by modern grids.
- **Delayed Attack Detection:** Without automated correlation and AI-based analytics, malicious activity can persist undetected for extended periods, increasing the potential for cascading operational impact.

1.4 System Objectives

The Smart Grid Cybersecurity Management Platform is guided by the following core objectives:

- To provide continuous, real-time monitoring of all smart grid devices, including smart meters, substations, and IoT sensors, across the distribution network.
- To detect cyber attacks, malware, and intrusions targeting both IT and OT segments of the smart grid with high accuracy and minimal false positives.
- To leverage artificial intelligence and machine learning for predictive threat detection and early identification of emerging attack patterns.
- To deliver real-time alerts and notifications to relevant stakeholders through a centralized, role-based security dashboard.
- To implement a quantitative risk scoring framework that prioritizes incidents based on severity, asset criticality, and potential operational impact.
- To enable automated, policy-governed incident response actions that reduce mean time to containment for confirmed threats.

1.5 Key Stakeholders

The successful design and operation of the platform depends on the coordinated involvement of a diverse group of stakeholders, each with distinct interests and responsibilities:

- To implement a quantitative risk scoring framework that prioritizes incidents based on severity, asset criticality, and potential operational impact.
- To enable automated, policy-governed incident response actions that reduce mean time to containment for confirmed threats.

Stakeholder	Role	Primary Interest
Utility Company Executives	Strategic decision-makers	Business continuity, regulatory compliance, and return on security investment
Grid Operators	Day-to-day management of grid operations	Uninterrupted power delivery and rapid awareness of operational disruptions
Cybersecurity Analysts	Threat detection and investigation	Accurate, timely detection of attacks and actionable forensic data
Network Administrators	Management of IT/OT network infrastructure	Network stability, segmentation integrity, and traffic visibility
Government Regulatory Bodies	Oversight and enforcement	Verifiable compliance with critical infrastructure protection standards
Field Engineers	On-site maintenance of grid equipment	Accurate device status information and safe remote intervention capability
Customers	End consumers of electricity	Reliable service and protection of personal consumption data
System Administrators	Platform maintenance and configuration	System uptime, scalability, and ease of administration
AI Security Team	Development and tuning of detection models	Access to quality data and feedback loops for model improvement
Emergency Response Team	Incident containment and recovery	Rapid, coordinated response workflows during active incidents

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2. Functional and Non-Functional Requirements

2.1 Functional Requirements

The following table enumerates the core functional requirements of the Smart Grid Cybersecurity Management Platform, organized by module:

ID	Module	Requirement
FR-01	Threat Detection	The system shall continuously analyze network and device telemetry to detect known and unknown cyber threats.
FR-02	Intrusion Detection	The system shall monitor IT and OT network segments to detect unauthorized access attempts and lateral movement.
FR-03	User Authentication	The system shall require multi-factor authentication for all users accessing the platform.
FR-04	Incident Response	The system shall allow authorized users to trigger predefined response playbooks for confirmed incidents.
FR-05	Device Monitoring	The system shall maintain a real-time status inventory of all connected grid devices.
FR-06	Network Monitoring	The system shall capture and analyze network flow data across substations and field networks.
FR-07	Risk Assessment	The system shall calculate a dynamic risk score for each detected event based on severity and asset criticality.
FR-08	Compliance Monitoring	The system shall track adherence to configured regulatory and industry security standards.
FR-09	Reporting	The system shall generate scheduled and on-demand security and compliance reports.
FR-10	Alert Management	The system shall generate, prioritize, and route real-time alerts to appropriate stakeholders.
FR-11	Audit Logs	The system shall maintain immutable audit logs of all user and system actions.
FR-12	AI Threat Prediction	The system shall apply machine learning models to forecast potential attack scenarios.
FR-13	Role Based Access	The system shall enforce role-based access control across all platform modules.

2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes the platform must exhibit to be operationally viable within a critical infrastructure environment:

Attribute	Description
Performance	The system shall process and analyze telemetry from at least 100,000 devices with alert generation latency under two seconds.
Reliability	The system shall maintain accurate detection performance with a false positive rate below industry-accepted thresholds.
Availability	The system shall achieve 99.9% uptime, reflecting the critical nature of grid security operations.
Scalability	The system shall scale horizontally to accommodate growth in device count and renewable energy integration.
Security	The system itself shall be hardened against compromise, including secure coding practices and regular penetration testing.
Maintainability	The system shall be built on modular components that can be updated independently without full system downtime.
Interoperability	The system shall support standard industrial protocols such as DNP3, Modbus, and IEC 61850.
Usability	The dashboard shall present complex security data through intuitive, role-appropriate visualizations.
Compliance	The system shall support configurable compliance frameworks aligned with critical infrastructure protection standards.
Auditability	All system actions shall be logged in a tamper-evident manner suitable for regulatory audit.
Data Privacy	Customer consumption data shall be anonymized or pseudonymized wherever operationally feasible.
Disaster Recovery	The system shall support automated backup and recovery with a recovery time objective of under four hours.

3. Actors and Use Cases

The platform involves a combination of primary actors, who directly interact with the system to perform operational tasks, and secondary actors, which support the system through automated services and external integrations.

Actor	Type	Key Use Cases
Grid Operator	Primary	Monitor grid status, acknowledge alerts, coordinate operational response
Cybersecurity Analyst	Primary	Investigate threats, review logs, tune detection rules
System Administrator	Primary	Manage users, configure platform, maintain system health
Utility Executive	Primary	View compliance and risk reports, oversee strategic decisions
Field Engineer	Primary	Verify physical device status, execute on-site remediation
Government Regulator	Secondary	Review compliance reports and audit trails
AI Engine	Secondary	Generate predictive threat models and risk scores
Authentication Service	Secondary	Validate user credentials and enforce access control
Notification Service	Secondary	Deliver alerts via email, SMS, and dashboard channels

4. Software Requirement Specification (SRS)

4.1 Introduction and Scope

This section defines the boundaries of the Smart Grid Cybersecurity Management Platform, clarifying the capabilities that fall within the project's delivery scope and those that are explicitly excluded.

In Scope

- Real-time monitoring and threat detection across smart meters, substations, and IoT field devices.
- AI-based threat prediction, risk scoring, and automated alert generation.
- Centralized security dashboard with role-based access for all stakeholder categories.
- Automated incident response workflows for predefined, low-risk containment actions.
- Compliance monitoring and reporting against configurable regulatory frameworks.

Out of Scope

- Physical replacement or hardware upgrade of legacy substation equipment.
- Direct control of generation dispatch or energy market trading functions.

4.2 Key Use Case Descriptions

Use Case	Actor	Main Flow	Postcondition / Exception
UC1 Monitor Grid Devices	Grid Operator	Operator views live device status across substations and meters, drilling into flagged devices.	Health view confirmed; unreachable devices raise a connectivity alert.
UC2 Detect Cyber Attack	Cybersecurity Analyst, AI Engine	System analyzes traffic and behavior; anomalies matching attack indicators generate a security event.	Threat alert logged; false positives dismissed with model feedback.
UC3 Predict Threat	AI Engine	Model evaluates current patterns against learned indicators to score emerging threats.	Prioritized predictions shown; low-confidence results flagged for review.
UC4 Generate Incident Alert	Notification Service	System compiles event details and dispatches alerts via dashboard, email, and SMS.	Stakeholders notified in time; failed channels retried and logged.
UC5 Respond to Incident	Cybersecurity Analyst	Analyst selects a response playbook and executes or authorizes automated containment.	Threat contained; failed automation escalates to a field engineer.
UC6 Generate Security Report	Utility Executive, Admin	User selects scope and framework; system compiles metrics into a formatted report.	Report available for audit; incomplete data sections are flagged.

4.4 External Interface Requirements

- User Interface: A responsive, web-based dashboard with role-specific views for operators, analysts, executives, and administrators, visualizing device status, alerts, and compliance posture.

- **Hardware Interface:** Interfaces with substation IEDs, RTUs, smart meters, and network sensors via standard industrial gateways without proprietary hardware changes.
- **Software Interface:** Exposes and consumes RESTful APIs to integrate with existing SCADA, meter data management, and asset management systems.
- **Communication Interface:** Secure communication over TLS-encrypted channels and industrial protocols including DNP3, Modbus TCP, and IEC 61850.
- **Cloud Integration:** Deployment on cloud infrastructure with secure, encrypted connectivity to on-premises grid control centers.
- **IoT Integration:** Lightweight, low-bandwidth protocols such as MQTT for resource-constrained IoT field devices.

4.5 System Features

- **AI Threat Detection:** Continuous, model-driven analysis of network and device behavior to identify malicious activity.
- **Machine Learning Prediction:** Forecasting of likely attack scenarios based on historical patterns and current indicators.
- **SIEM Integration:** Ingestion and correlation of security event data from third-party Security Information and Event Management systems.
- **Real-Time Monitoring:** Live telemetry ingestion from grid devices with sub-second refresh on the operational dashboard.
- **Security Dashboard:** Unified, role-based visualization of grid-wide security posture and active incidents.
- **Incident Management:** End-to-end tracking of incidents from detection through resolution and post-incident review.
- **Risk Analysis:** Quantitative scoring of assets and events to prioritize response effort.
- **Compliance Reports:** Automated generation of reports aligned with configurable regulatory frameworks.
- **Role-Based Access:** Granular permission control ensuring users access only the functions relevant to their role.
- **Alert Notifications:** Multi-channel delivery of real-time alerts to appropriate stakeholders.

5. Assumptions and Constraints

5.1 Assumptions

- All grid devices are equipped with, or can be retrofitted with, network connectivity sufficient to report telemetry to the platform.
- Utility companies will provide the necessary network access and firewall configurations for platform integration with existing SCADA systems.
- Historical security event data of sufficient volume and quality is available to train initial AI detection and prediction models.

- Stakeholders will receive adequate training to interpret dashboard outputs and execute recommended response actions.
- Regulatory compliance frameworks referenced by the platform remain relatively stable during the initial deployment period.
- Cloud infrastructure providers used for deployment maintain industry-standard physical and network security controls.

5.2 Constraints

Category	Description
Technical	The platform must interoperate with legacy SCADA equipment using older, less secure communication protocols.
Operational	Deployment must not disrupt continuous grid operations, requiring careful phased rollout planning.
Economic	Budget constraints limit the scope of hardware replacement, favoring software-based security enhancements.
Legal	The platform must comply with data protection regulations governing customer consumption data.
Time	Implementation must align with the utility's regulatory reporting deadlines for cybersecurity readiness.
Infrastructure	Rural or remote substations may have limited bandwidth, constraining real-time data transmission.

6. Requirement Validation

6.1 Completeness Check

A structured review confirmed that every functional area from the scenario analysis — device monitoring, threat detection, risk scoring, incident response, and compliance reporting — maps to at least one functional requirement, and that all stakeholder interests are addressed through defined use cases with no orphaned business need identified.

6.2 Consistency and Ambiguity Check

Requirements were checked for internal consistency to avoid contradictory expectations on the same component, and ambiguous terms were replaced with measurable language — for example, vague 'fast' alerting was refined into an explicit two-second latency target, and 'secure' communication was tied to specific protocol and encryption requirements.

7. Conclusion

The Smart Grid Cybersecurity Management Platform addresses a pressing need in the energy sector: securing an increasingly digital, distributed power infrastructure against a sophisticated threat landscape. By consolidating device monitoring, intrusion detection, AI-based threat prediction, and automated incident response into one platform, it resolves

the fragmentation and visibility gaps common in current utility security operations. Commercially, it reduces the cost of manual monitoring, minimizes downtime from undetected incidents, and strengthens regulatory and customer confidence, while AI integration gives it a durable edge in adapting to novel attack techniques beyond static, signature-based detection.

Beyond the individual utility, the platform strengthens the resilience of critical national infrastructure and reduces systemic risk from cascading grid failures. Its modular, cloud-native architecture supports future scalability as smart meters, IoT sensors, and renewable integration continue to grow, while faster detection and containment improve grid reliability and operational efficiency. Its centralized dashboard fosters coordination among operators, analysts, and field engineers — positioning utility companies to meet future security demands as part of the broader digital transformation of the energy sector.

Code of Conduct

I declare that the analysis, requirements, and content presented in this report represent my own original work, developed in accordance with academic integrity guidelines. All requirement statements, use cases, and diagrams have been independently authored for the stated project title, and any external references have been appropriately acknowledged.

Signature of the student : N.MADHUSUDHAN

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